

# ECONOMIC AWARENESS & SOVEREIGN MONETARY ARCHITECTURE

A TIA / Genisys-Inspired Architecture for the Human Economy, the  
Productive Economy and Proof-Bound Monetary State

## ARCHITECTURE THESIS

SunKey separates the ability to understand economic reality from the authority to change monetary state. Federated sources may observe, normalize, correlate and verify. Canonical claims may be valued. Only a governed protocol path may finalize supply-changing monetary state.

**Economic facts outside the chain. Monetary truth inside the chain.**

SUNREY TECHNICAL WHITE PAPER | SEPTEMBER 2026

Public architecture paper. Conceptual design; regulated capabilities, production monetary parameters and claims of technical novelty remain subject to legal, security, governance and jurisdiction-specific review.

## FOUNDATIONAL THESIS

# Abstract

Modern economic systems increasingly depend on information that originates outside their settlement layer: credentials, research records, measurements, compute receipts, production telemetry, logistics events, identity attestations and other evidence. The design problem is not merely how to ingest more data. It is how to make distributed economic facts legible to a sovereign monetary system without allowing any data source, database, AI model, oracle, exchange or application service to become a monetary authority.

SunRey proposes a proof-bound architecture in which an Economic Awareness Fabric federates heterogeneous sources, preserves provenance and rights, resolves entities, evaluates source independence and constructs evidence-sufficient propositions. Those propositions become Canonical Economic Claims with durable event identity and anti-double-counting controls. Human and productive valuation layers may then interpret eligible claims, but valuation remains deliberately separate from monetary policy and market price.

## Economic Awareness Outside the Chain, Monetary Truth Inside the Chain

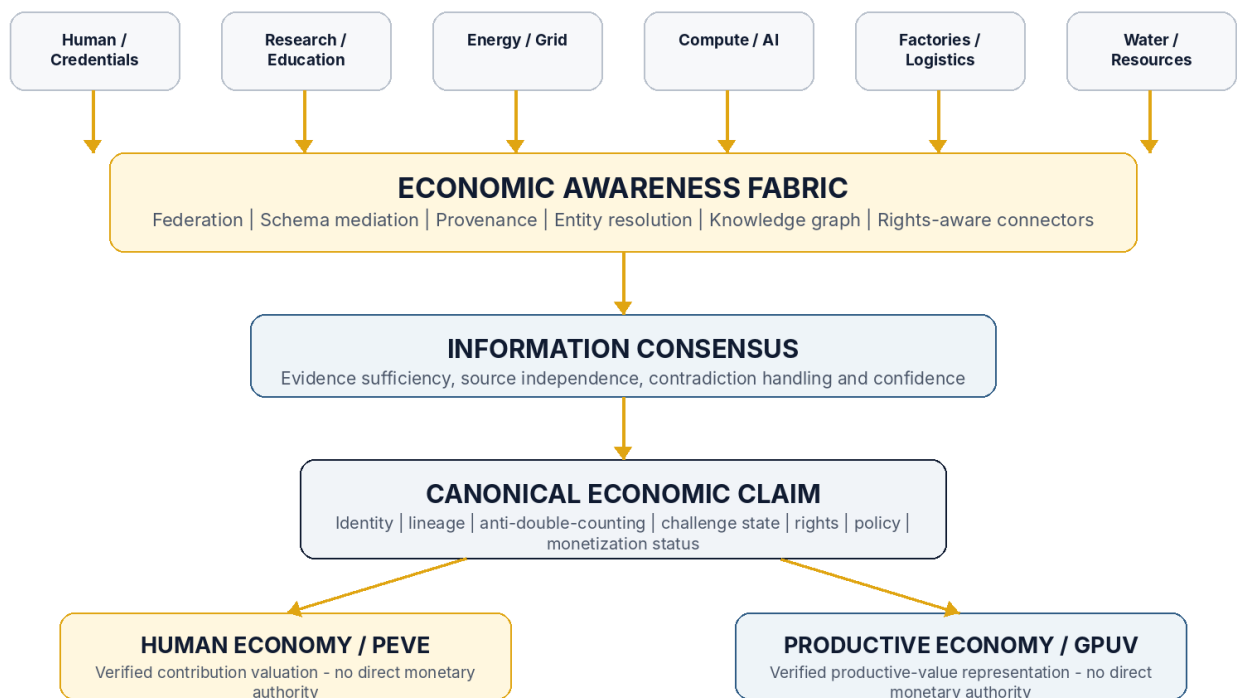


Figure 1. SunRey places federated economic awareness outside the monetary core and preserves a narrow, governed path into sovereign state.

### Core invariant

Information may support an economic claim. A claim may support a valuation. A valuation may support a monetary proposal. None of those layers can independently create money.

## EXECUTIVE SUMMARY

# Economic awareness without monetary authority drift

SunRey begins from a premise that conventional token architectures often begin too late. They model an asset, a balance and a transfer, then connect external information afterward through oracles or application logic. A system intended to coordinate human contribution and autonomous productive output needs to begin earlier: what economic event occurred, what evidence supports it, who or what did it involve, what rights govern its use, whether it is unique, how it should be interpreted, and what authority - if any - permits it to influence monetary state.

- The Economic Awareness Fabric (EAF) is a federated information and evidence layer, not a minting or monetary service.
- Information Consensus answers whether authorized evidence is sufficient to support an economic fact; Monetary Consensus answers whether a proposed protocol state transition is valid and final.
- Canonical Economic Claims create one durable economic identity for an event across many observations, enabling lineage, challenge state and anti-double-counting.
- The Economic Knowledge Graph connects entities, events, assets, contributions and evidence while preserving source lineage and purpose constraints.
- PEVE and GPUV are valuation layers for human and productive contribution respectively; neither determines native-asset supply or exchange price.
- Five proof commitments - Transaction Root, Monetary State Root, Evidence Root, Rights Root and Policy Root - bind monetary state to the context that justified it.
- AI may classify, summarize, correlate and propose; it remains subordinate to rights, policy, governance and protocol finality.

The architectural ancestry is deliberately limited. DARPA's early-2000s TIA/Genisys research demonstrated serious interest in federated data access, schema abstraction, relationship discovery and privacy-control mechanisms. The historical mission and governance controversies are not inherited. SunRey uses the reference as a systems-engineering lesson while inverting the trust model toward purpose limitation, minimization, user-linked rights and explicit separation between information and sovereign monetary authority.

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## 1 | PROBLEM DEFINITION

# Economic reality is not the same thing as monetary state

A sovereign economic protocol must interact with reality it does not control. Power generation occurs in physical infrastructure. Compute jobs execute in external clusters. Research is published through institutions. Credentials are issued by universities and employers. Logistics events are recorded by carriers. Human permissions can be granted and revoked. Prices form in markets. None of these systems should automatically become part of the monetary authority simply because their data is useful.

The core systems problem is therefore one of authority separation. The platform needs enough awareness to determine whether an economic event is sufficiently supported, yet it must prevent evidence collection from collapsing into money creation. It also needs enough semantic structure to distinguish a genuine new event from a duplicate, a component from an aggregate, a claim from a source record, and a valuation from a monetary decision.

### Problem statement

How can a system federate distributed evidence about human and productive economic activity, determine factual sufficiency, preserve rights and provenance, prevent replay or double-counting, and still keep monetary finality under a narrow, governed protocol authority?

SunRey answers by inserting explicit layers between external observations and monetary state. Each layer is deliberately capable of less authority than the layer below it. This is not only a data architecture; it is an authority architecture.

# TIA was an experimental prototype network, not simply a giant database

The historical reference point is the Defense Advanced Research Projects Agency (DARPA) Total Information Awareness program, renamed Terrorism Information Awareness in 2003. A March 27, 2003 DARPA director testimony described TIA as an experimental, multi-agency prototype network intended to integrate collaboration and decision support, language translation, and data-search and pattern-recognition technologies [1]. The May 20, 2003 report to Congress likewise described TIA as a program-of-programs whose component technologies would be integrated and stress-tested in prototype configurations [2].

That distinction matters. The useful architectural lesson is not the creation of one universal database. It is the orchestration of heterogeneous information systems through a common architecture, with shared interfaces and analytical services that could reason across sources while leaving source systems distinct.

The historical program was also inseparable from controversy over privacy and civil liberties. The DoD Inspector General later noted both the existence of privacy-protection research and the need for more comprehensive privacy impact analysis before TIA-type technology proceeded into operational use [4]. DARPA's later institutional history characterizes TIA as highly controversial and terminated amid that controversy [5].

## Publication boundary

SunRey uses TIA / Genisys only as architectural ancestry. It does not inherit the counterterrorism mission, centralized surveillance logic or any premise that information access itself confers operational authority.

## Federation, mediation and schema abstraction

Genisys is the component most relevant to SunRey's information architecture. The May 2003 congressional report states that Genisys aimed to develop a federated database architecture allowing analysts to obtain answers across multiple databases without needing to know where information resided or how each source was internally structured [2]. The same report lists an abstract-schema milestone, while DARPA budget material describes work intended to simplify use of databases by reducing dependence on their internal schemas [3].

The modern systems lesson is straightforward: interoperability should be expressed through mediation and canonical semantics rather than by forcing every source into one physical repository. In SunRey, that becomes the Economic Observation Envelope and the Economic Awareness Fabric. A grid operator, credential issuer, compute provider and logistics platform do not need to share one storage model; they need to expose evidence that can be normalized into common economic semantics without surrendering source lineage.

This separation supports replaceability. A provider connector can change without changing the meaning of an economic claim. Likewise, a federation engine can be replaced without changing monetary semantics. The monetary core therefore depends on canonical claims and proof commitments, not on the continued operation of a particular data warehouse or query system.

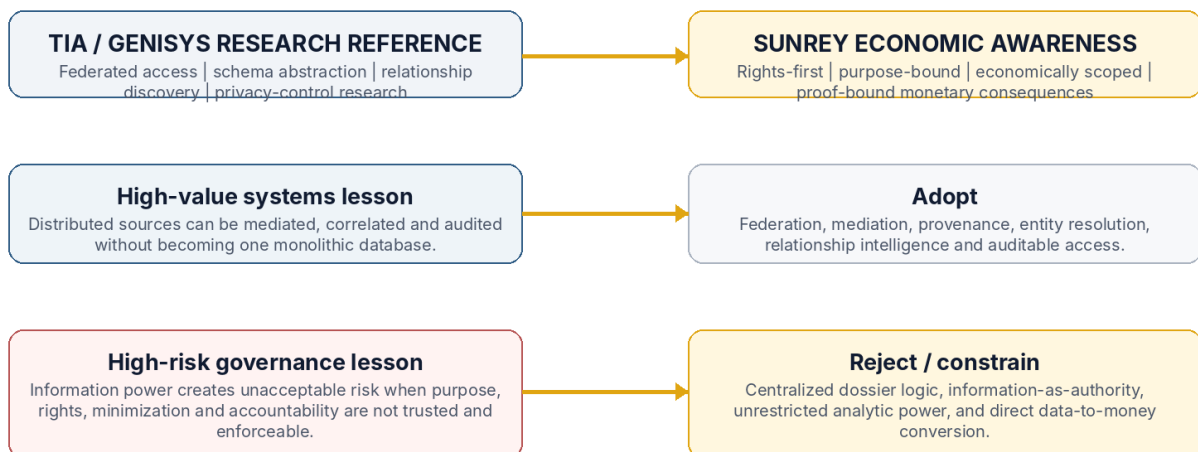
## 2.3 | PRIVACY &amp; RELATIONSHIP INTELLIGENCE

## Privacy controls and relationship discovery were explicit research problems

Genisys Privacy Protection was not incidental. The 2003 report to Congress described research into role-based rules, inference control, automated access control and immutable audit, along with anonymization and controls intended to expose identity only when further authorization was justified [2]. The DoD Inspector General similarly summarized Genisys Privacy Protection as work on access control, inference control and analyst tracking, while recommending a more comprehensive privacy impact process for TIA-type technology [4].

Evidence Extraction and Link Discovery (EELD) addressed a complementary problem: extracting relationships among people, organizations, places and things from unstructured information and connecting related evidence for human analysis [2]. SunRey does not reuse the investigative mission. It reuses the systems concept that economically meaningful relationships often emerge only when multiple records are normalized and linked to canonical entities and events.

### Architectural Ancestry, Mission Inversion



Same systems-engineering insight. Different mission, rights model and authority boundary.

Figure 2. SunRey adopts selected systems-engineering lessons while explicitly rejecting information-as-authority and centralized dossier logic.

## 2.4 | HISTORICAL LESSON

# The governance failure mode is part of the architecture lesson

The external historical fact-check supports four points that matter to the SunRey design. First, DARPA itself described TIA as an experimental prototype network, which corrects the oversimplified description of TIA as only a single giant database [1]. Second, Genisys explicitly pursued federated access and schema abstraction [2][3]. Third, privacy-protection research included access control, inference control, anonymization and immutable audit concepts [2][4]. Fourth, congressional action in 2003 eliminated funding for the majority of TIA components and directed termination of the Information Awareness Office; the DoD Inspector General documented that change and the surrounding privacy concerns [4].

The publication implication is not that historical TIA technology should be revived. It is that a system with powerful federation and relationship-awareness capabilities must treat governance as part of the technical design. Rights, purpose limitation, minimization, source licensing, auditability and narrow authority boundaries cannot be left as policy documents outside the architecture.

### Design lesson

High-value lesson: federation and relationship awareness. High-risk lesson: information power without sufficiently trusted rights, purpose and accountability creates unacceptable surveillance and control risk.

## 3 | SUNREY DESIGN INVERSION

## Economic awareness, not total information awareness

SunRey narrows the objective from broad awareness about a person or system to sufficient, authorized awareness of a specific economic proposition. A Human Economy flow should establish that a contribution occurred, that it is attributable to the correct canonical participant, that the relevant use is authorized, and that the same contribution has not already been consumed as a monetizable event. A Productive Economy flow should establish that useful output occurred, that source classes are sufficiently independent, that event boundaries are resolved, and that the same underlying output is not being counted repeatedly.

| Historical principle            | SunRey adaptation                                           | Decision   |
|---------------------------------|-------------------------------------------------------------|------------|
| Federated heterogeneous data    | Economic Awareness Fabric                                   | Adopt      |
| Schema mediation / translation  | Economic Observation Envelope                               | Adopt      |
| Relationship discovery          | Economic Knowledge Graph + entity resolution                | Adopt      |
| Pattern / evidence correlation  | Information Consensus + corroboration                       | Adopt      |
| Privacy controls / audit        | Rights, purpose, consent, policy receipts and roots         | Strengthen |
| Centralized dossier mentality   | Pseudonymous identity + minimization + computation-in-place | Reject     |
| Information access as authority | Information may support claims but cannot mint              | Reject     |
| Analytic system decides outcome | Governance + protocol consensus remain separate             | Reject     |

### Primary inversion

SunRey attempts to know enough to prove an economic claim - and no more than necessary - while preserving a governed, auditable boundary between evidence and monetary consequences.

## 4 | REFERENCE ARCHITECTURE

# A narrow monetary core surrounded by richer economic intelligence

The reference architecture is intentionally asymmetric. The information side is broad and extensible because new economic domains can be added over time. The monetary side is narrow because the number of components allowed to change native monetary state should remain small, explicit and governable.

## Economic Awareness Outside the Chain, Monetary Truth Inside the Chain

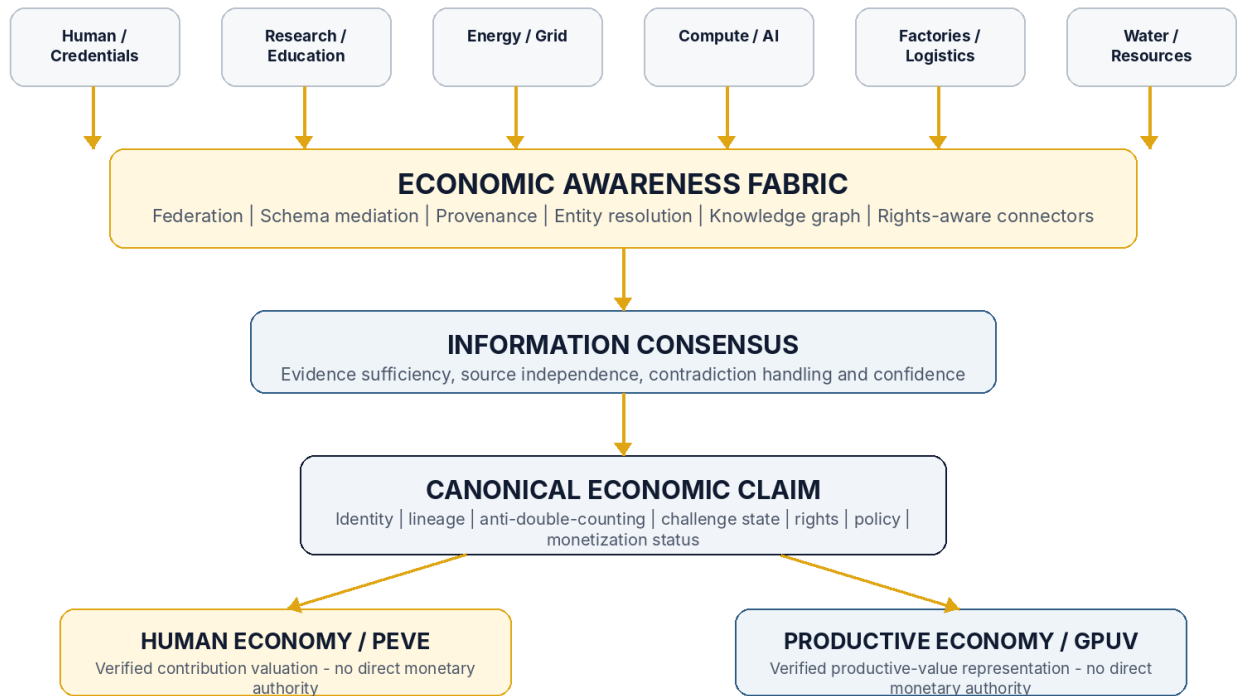


Figure 3. Reference architecture from external sources to canonical claims and domain-specific valuation.

The Economic Awareness Fabric may register providers, run connectors, normalize observations, preserve provenance, resolve entities, construct graph relationships, identify source dependencies, calculate confidence and propose verified facts or claims. The canonical claim layer then freezes a stable economic proposition with its evidence and policy context. Human and productive valuation engines interpret eligible claims. Monetary policy and protocol consensus remain downstream.

# The EAF is an economic evidence fabric, not a monetary service

The Economic Awareness Fabric (EAF) sits between external reality and the economic proof layer. It is the modern SunRey analogue to the useful federation concept in Genisys: providers remain heterogeneous and independently governed, while SunRey standardizes how their observations are represented, traced, compared and authorized.

- Provider registry: captures source identity, licensing status, domain, geography, authority class and operational health.
- Connector layer: retrieves or receives source records while preserving request, transport and transformation provenance.
- Normalization: converts provider-specific records into a canonical Economic Observation Envelope.
- Entity resolution: maps provider aliases and local identifiers to canonical people, organizations, assets, locations and events.
- Knowledge graph: expresses relationships among evidence, events, assets, contributors, rights and claims.
- Source-dependency analysis: identifies whether apparently independent endpoints share one upstream origin.
- Corroboration and contradiction analysis: evaluates support, disagreement, freshness and confidence before a claim progresses.

## Authority boundary

The EAF may observe, normalize, correlate, deduplicate and recommend. It cannot issue SunRey Coin, issue MoonRey Coin, set total supply, determine exchange price or approve monetary governance.

## 6 | ECONOMIC OBSERVATION ENVELOPE

## Normalize meaning without erasing provenance

A canonical observation envelope is the interface contract between heterogeneous providers and SunRey economic reasoning. Its purpose is not to force every source into the same raw schema. Its purpose is to preserve enough common structure that evidence can be compared, deduplicated, licensed and linked to canonical entities.

| Envelope field          | Purpose                                                                                        |
|-------------------------|------------------------------------------------------------------------------------------------|
| Observation identity    | Unique source record / event reference and ingestion identity.                                 |
| Source identity         | Provider, endpoint, upstream lineage, source class and authority category.                     |
| Subject / asset         | Canonical or unresolved person, organization, asset, machine, facility or location reference.  |
| Time                    | Observed-at, effective interval, publication time, freshness and timezone semantics.           |
| Measurement / assertion | What the source says occurred, including units, bounds and uncertainty.                        |
| Rights / license        | Permitted purposes, jurisdiction, retention, redistribution and commercialization constraints. |
| Provenance              | Transformations, signatures, hashes, attestations and chain of custody.                        |
| Quality state           | Verification level, contradiction state, confidence and challenge status.                      |

A source-specific payload can remain attached or retrievable outside the monetary chain. What becomes canonical is the economic meaning and proof context. This allows SunRey to add providers without rewriting core monetary logic and allows providers to be replaced without rewriting historical claims.

## 7 | SOURCE LINEAGE

## Corroboration must measure independence, not endpoint count

A common oracle failure mode is apparent redundancy without true independence. Five APIs can look like five sources while all five ultimately copy the same upstream database or market feed. Counting endpoints would overstate confidence and create a single-source dependency that is invisible to the monetary layer.

SunRey therefore treats source lineage as part of the evidence model. A source graph records upstream origin, transformation relationships, aggregation, publication chains and shared dependencies. Corroboration can then weight source classes rather than raw connector count. For productive claims, a direct meter, operator record and independent physical reference may provide stronger support than several commercial dashboards that repeat the same operator feed.

- Independent source class: evidence produced through a materially distinct observation or authority path.
- Derived source: evidence that transforms or republishes an upstream record.
- Correlated source: independent in ownership but not necessarily independent in underlying measurement.
- Authoritative source: a source that has recognized control or certification authority for a specific domain, without becoming monetary authority.
- Reference source: contextual evidence useful for plausibility checks but insufficient alone for monetary progression.

### Oracle resilience

The objective is not maximum data volume. It is sufficiently independent, authorized evidence for the economic proposition being evaluated.

# Factual sufficiency is a separate decision plane

## Two Consensus Planes, Two Different Questions

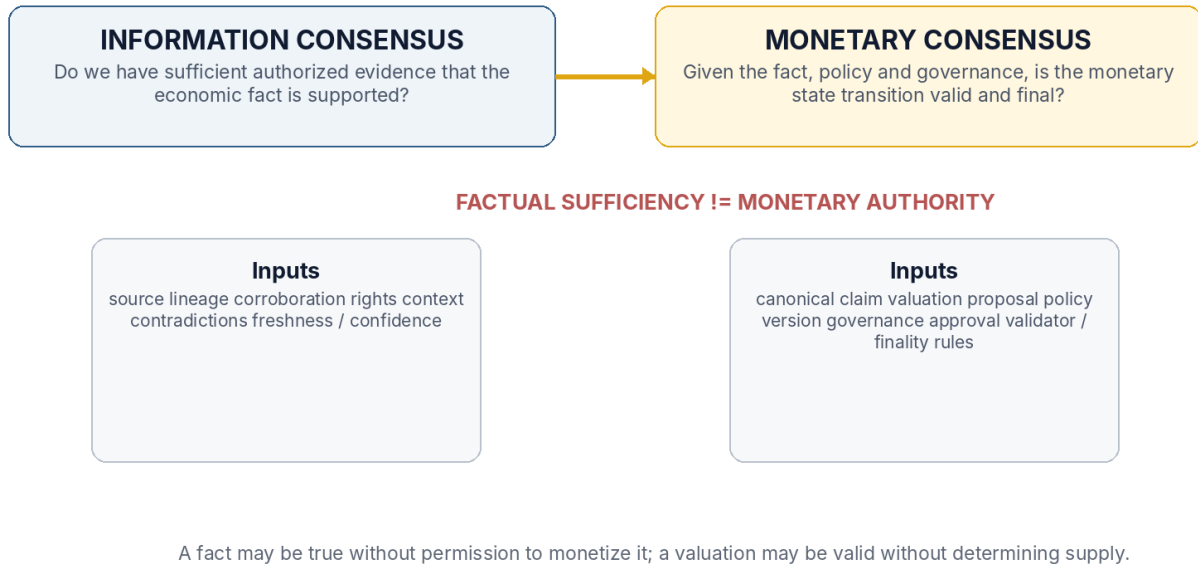


Figure 4. Information Consensus and Monetary Consensus answer different questions and must not collapse into one mechanism.

Information Consensus is SunRey terminology for an evidence-sufficiency mechanism. It does not imply that every economic fact must be decided by the same Byzantine-fault-tolerant validator process used for monetary finality. Domain policies may use deterministic rules, threshold attestations, confidence bands, challenge periods, trusted authorities or other methods appropriate to the economic object.

Its output is a supported fact or claim candidate with an explicit confidence, evidence set and policy version. The output still has no minting authority. A source can be correct without being authorized for a particular purpose. A claim can be well-supported but ineligible for monetary use. A valuation can be technically valid but not authorized to influence supply.

## 9 | CANONICAL ECONOMIC CLAIMS

# A durable economic proposition between raw evidence and monetary logic

The Canonical Economic Claim is the central object that stabilizes economic meaning. It represents a durable proposition such as a verified research contribution, a certified compute job, a measured generation event or another economically relevant occurrence. The claim carries canonical entity and event identity, evidence lineage, rights references, policy references, duplicate fingerprints, challenge state and monetization status.

## Canonical Economic Claim and Anti-Double-Counting

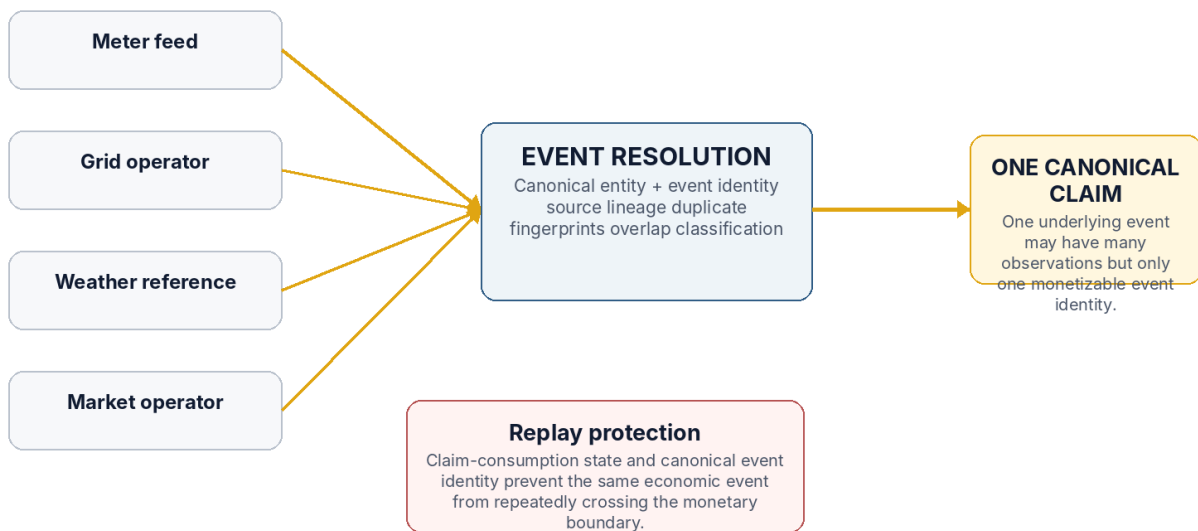


Figure 5. Multiple observations can support one event; the canonical claim prevents observations from becoming multiple monetizable events.

This layer creates a clean separation between evidence and economic consequences. Raw records can be corrected, revoked, challenged or superseded while the canonical claim maintains an append-only history of what was known, under what policy and with what authorization at each point in time.

## Economic replay is a monetary integrity problem

A proof-bound monetary system cannot treat every record as a new economic event. Duplicate data, aggregates, component-level reports, republished research, multiple wallets and repeated oracle observations can all create false multiplicity. Without canonical event identity, the same underlying economic activity can be wrapped repeatedly and cross the monetary boundary more than once.

SunRey therefore treats event reconciliation as domain-aware. In a productive economy, capacity is not production; stock is not flow; daily and monthly aggregates cannot be naively summed; and parent and child assets cannot both receive full attribution for the same output. In a human economy, multiple publication indexes can refer to one research contribution, and multiple wallets controlled by one participant do not create multiple people or contributions.

| Resolution state         | Monetary treatment                                                          |
|--------------------------|-----------------------------------------------------------------------------|
| Exact duplicate          | Same record or cryptographic content; no new event.                         |
| Same-event corroboration | Independent evidence supports the same underlying event.                    |
| Partial overlap          | Records share some output or time interval and require bounded allocation.  |
| Aggregate-of             | One record summarizes multiple underlying events.                           |
| Component-of             | One record is a component of a larger parent event.                         |
| Distinct event           | Evidence supports a separate event identity.                                |
| Unresolved               | System cannot safely determine uniqueness; monetary progression is blocked. |

### Claim consumption

Once an eligible claim is used in a supply-changing process, its monetization state should be durably recorded so replay attempts are rejected or explicitly governed.

## Economic relationships become first-class evidence

The Economic Knowledge Graph provides a semantic layer across observations, canonical entities, claims, rights and productive relationships. Its purpose is not to become a second ledger. It gives the EAF a way to reason about relationships that are economically relevant but distributed across source systems.

- Identity relationships: person-to-credential, organization-to-issuer, machine-to-controller, wallet-to-economic identity.
- Asset relationships: unit-to-facility, facility-to-grid, vehicle-to-fleet, compute node-to-cluster.
- Event relationships: component-of, aggregate-of, predecessor, derived-from, same-event-corroboration.
- Rights relationships: grantor, grantee, purpose, scope, jurisdiction, expiration, revocation and downstream-use constraints.
- Evidence relationships: source-to-claim, source-dependency, contradiction, challenge and supersession.
- Economic relationships: contribution-to-valuation, productive event-to-GPUV, claim-to-monetary-proposal and proposal-to-finalized-state.

High-impact entity merges should not be silently determined by an AI model. Automated resolution may propose links, but deterministic thresholds, authoritative evidence or human review are required when a merge could change ownership, attribution, uniqueness or monetary eligibility.

## Human contribution is not human worth

The Human Economy branch uses the shared claim and evidence infrastructure while maintaining its own ontology. SunRey is designed to value defined, authorized contributions rather than assign an intrinsic score to a person. Sensitive attributes such as health, DNA, psychology, age, location or social relationships are not economic contributions merely because information exists about them.

Where sensitive information participates in an authorized research or computation event, the monetizable object is the specific contribution or computation event under its applicable rights, not the person's underlying characteristic. Pseudonymous human economic identity can preserve canonical attribution while keeping login identity, wallet identity and governance identity distinct.

- Contribution Attestation Mesh: contribution-specific evidence from credential issuers, employers, research institutions, signed compute receipts or consented data-use receipts.
- Sybil resistance: contribution uniqueness attaches to canonical participant/event identity rather than wallet count.
- Purpose-bound rights: evidence can be valid for research but not automatically valid for monetary use.
- PEVE: a valuation layer that interprets an eligible contribution under a versioned methodology without becoming monetary authority.

## Productive output requires canonical assets, events and source-aware oracles

The Productive Economy branch applies the same proof discipline to physical and digital output. Productive assets can include power plants, data centers, compute clusters, factories, mines, farms, logistics hubs, vehicles, telecom infrastructure and water systems. Provider-specific aliases resolve to canonical assets with explicit parent-child hierarchies so that facility-level and unit-level output is not automatically counted twice.

The Productive Oracle Mesh is source-lineage-aware. Energy evidence might combine direct metering, operator or grid records and independent reference data. Compute evidence may combine workload receipts, cluster telemetry and energy/resource evidence. Manufacturing evidence may combine enterprise production records with logistics or resource flows. The important property is not the number of endpoints but the independence and authority characteristics of the evidence classes.

GPUV - Gross Productive Unit Value in the SunRey architecture - is a standardized productive-value representation downstream of verified productive contribution and upstream of monetary policy. It remains distinct from MoonRey quantity and MoonRey market price.

## 14 | VALUATION SEPARATION

## Valuation, monetary policy and market price are separate planes

One of the architecture's most important protections is the refusal to collapse economic measurement into token supply. Human contribution and productive output may each require domain-specific valuation methodologies, but neither valuation engine can independently determine native-asset quantity. Likewise, an exchange price cannot become a valuation oracle merely because it is observable.

| Plane                        | Function                                                               | May create supply?       |
|------------------------------|------------------------------------------------------------------------|--------------------------|
| Observed economic reality    | Measurements, attestations, receipts and evidence about what occurred. | No                       |
| Information Consensus        | Evidence sufficiency for a fact or canonical claim.                    | No                       |
| PEVE / GPUV                  | Versioned economic valuation representation.                           | No                       |
| Monetary policy / governance | Rules determining whether and how eligible value may influence supply. | Proposal / authorization |
| Monetary Consensus           | Validation and finality of authorized monetary transitions.            | Finalization path        |
| Exchange market price        | Price discovered among market participants.                            | No                       |

### Four-plane productive separation

Physical measurement != productive value (GPUV) != monetary supply policy != market price. The same separation applies to human contribution: contribution evidence != PEVE != SunRey quantity != market price.

## Every upstream system is explicitly denied minting authority

The shared issuance firewall is the core monetary-safety model. It defines negative authority as clearly as positive authority. Raw personal data cannot mint. An Economic Observation cannot mint. A Verified Economic Fact cannot mint. A Canonical Economic Claim cannot mint. PEVE and GPUV cannot mint. An oracle, AI agent, API, frontend, exchange, provider database or knowledge graph cannot mint.

- Upstream systems may create evidence, calculations, classifications or proposals.
- Rights and purpose controls determine whether evidence may be used for a particular economic purpose.
- Valuation engines interpret eligible claims under versioned methodologies.
- Monetary policy determines whether an eligible valuation can form a supply proposal.
- Governance provides required approvals and policy authority.
- The canonical protocol validates the transition, rejects replay and finalizes state under its consensus rules.

### **Non-negotiable boundary**

Intelligence can inform monetary decisions. Intelligence - human or artificial - is not sovereign monetary authority.

## Finality applies to authorized state transitions, not to raw information

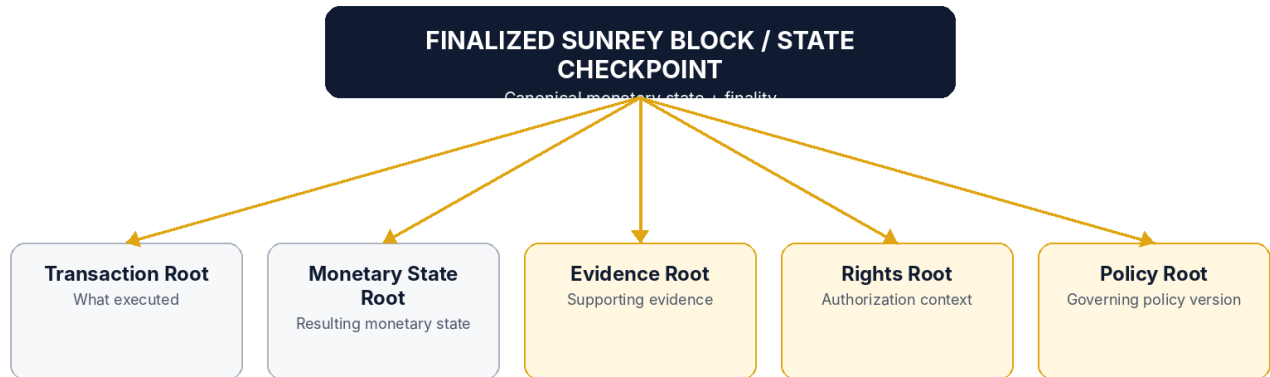
Monetary Consensus is the finality plane for protocol-native economic state. It evaluates a transaction or supply-changing proposal against canonical state, monetary policy, governance approvals, consumed-claim state, signature and validator rules, and other consensus-critical invariants. It should not independently decide whether an external economic fact is true; that question belongs to the information and claim layers.

The distinction limits the consensus surface. Validators need deterministic inputs representing the authorized economic consequence and its proof commitments, not unrestricted access to every external database. This reduces privacy exposure, avoids making validator nodes general-purpose data-integration systems, and makes consensus-critical execution easier to reproduce and audit.

A validator can therefore agree that a proposed state transition is invalid even when the underlying economic fact is true - for example because the policy version is ineligible, the rights authorization does not cover monetary use, the claim has already been consumed, or the required governance threshold was not met.

# Five roots bind monetary state to why it changed

## Proof-Bound Monetary State: Five Cryptographic Commitments



The roots are cryptographic commitments to context, not a requirement to publish sensitive source data on-chain.

Exact commitment primitive (Merkle tree or equivalent) is an implementation decision; the architectural invariant is proof linkage.

Figure 6. Proof-bound state links monetary finality to execution, evidence, rights and policy context without placing raw private data on-chain.

The five-root model is a conceptual commitment structure. Transaction Root commits to what executed. Monetary State Root commits to the resulting protocol state. Evidence Root commits to the supporting evidence set or proof package. Rights Root commits to the authorization context. Policy Root commits to the exact version of the governing methodology and rules.

The architecture does not require the chain to publish private source material. The roots can commit to external or encrypted evidence packages that remain access-controlled. Historical evidence can later be challenged, permissions can expire or be revoked for future use, and policies can change without rewriting finalized history. New states are represented as subsequent append-only events.

## A true fact can still be unauthorized for a particular use

SunRey separates truth from permission. RightsGrant, ConsentGrant, PurposeAuthorization and provider-license controls determine whether a piece of evidence may be used for contribution verification, research, valuation, model training, commercial analytics or monetary proposal. A purpose authorized for research does not automatically authorize monetization. A provider license that permits display may not permit model training or resale.

Privacy therefore becomes an architectural property rather than a downstream compliance label. Sensitive source data can remain encrypted or in a user-linked vault; computation-in-place can produce narrowly scoped outputs; identity can remain pseudonymous where legal and appropriate; access can be purpose-bound and revocable; and audit records can preserve who requested what under which authority.

- Data minimization: collect and expose only what the economic claim requires.
- Purpose limitation: authorize a specific use rather than general access.
- Separation of identity planes: login, wallet, economic and governance identity do not need to be identical.
- Computation-in-place: prefer verified outputs when raw records need not move.
- Append-only accountability: preserve decisions, policy versions and access evidence without rewriting history.

## AI is an analyst, not a sovereign

AI can materially improve the Economic Awareness Fabric. It can classify incoming records, suggest entity matches, detect anomalies, summarize contradictions, extract relationships from unstructured documents, explain confidence, propose claim boundaries and help users understand financial or economic implications. Those are powerful analytical roles.

The architecture deliberately denies AI the authority to silently merge high-impact identities, fabricate evidence, override rights, activate monetary policy, authorize issuance, elevate its own permissions or finalize monetary state. Agent capabilities should be scoped, revocable and purpose-bound. High-consequence operations require deterministic controls, signed authorization or human governance as defined by policy.

### AI safety principle

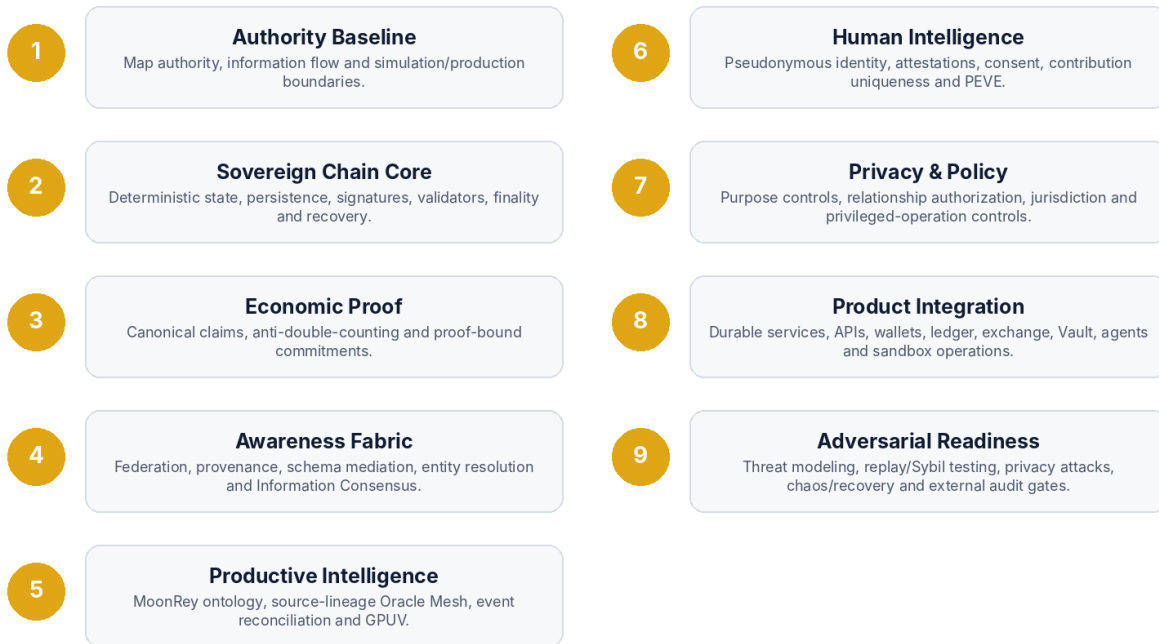
AI may help determine what evidence means. It does not determine what authority the evidence has.

This division also improves model replaceability. The system can change AI providers or models without changing the economic constitution. A new model may improve extraction or reasoning, but it cannot acquire additional monetary powers simply because it performs better.

## Nine controlled development phases

The architecture was developed in nine sequential phases. The sequencing is intentional: authority boundaries are defined before state becomes durable; canonical chain behavior is established before economic proof is attached; shared information infrastructure is established before human and productive domains are specialized; and product integration occurs only after privacy and policy boundaries are explicit.

### Nine Architecture Development Phases



The phases describe architectural sequencing and validation boundaries - not an internal coding workflow or deployment claim.

Figure 7. Architecture development is expressed as nine validation phases rather than an internal prompt or tool workflow.

Each phase should end with explicit exit criteria, regression testing and evidence that no component has acquired authority outside its intended domain. This discipline is particularly important in AI-assisted development because the most dangerous defect is often not a syntax error but authority drift: a convenient service, database field or automation gradually becomes able to do something reserved for governance or consensus.

## Use mature components behind SunRey-owned economic interfaces

The architecture does not depend on a preserved TIA or Genisys software stack. Historical programs are reference points, not implementation dependencies. Modern open-source components can supply federation, eventing, provenance, graph, policy, identity, privacy and consensus functions so long as each remains behind SunRey-owned domain interfaces and none becomes a source of canonical coin supply.

| Function                   | Illustrative substrate                | Role                                                                                                   |
|----------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------|
| Federated queries          | Trino or equivalent                   | Query approved heterogeneous sources in place.                                                         |
| Event streaming            | Apache Kafka or equivalent            | Durable high-throughput economic events and isolated consumers.                                        |
| Dataflow / provenance      | Apache NiFi or equivalent             | Connector orchestration and dataflow provenance outside consensus-critical execution.                  |
| Economic graph             | Apache AGE / PostgreSQL or equivalent | Graph relationships close to durable economic state while keeping graph projections non-authoritative. |
| Policy-as-code             | Open Policy Agent or equivalent       | Fail-closed, versioned purpose/jurisdiction decisions.                                                 |
| Relationship authorization | OpenFGA or equivalent                 | Fine-grained authorization among users, agents, resources and governance roles.                        |
| Identity federation        | Keycloak or equivalent                | OIDC/OAuth, MFA and identity brokering with separation from economic identity.                         |
| Privacy analytics          | OpenDP or equivalent                  | Aggregate analysis where exact individual monetary state is not required.                              |
| BFT finality               | CometBFT or equivalent                | Mature validator consensus around deterministic SunRey monetary state.                                 |

### Substrate rule

Technology choices are replaceable. Economic semantics, authority boundaries and canonical state are not.

## The highest-risk attacks target authority and economic uniqueness

| Failure mode        | Risk                                                                                       | Primary controls                                                                                 |
|---------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Authority drift     | A database, exchange, admin endpoint or AI agent gains an unintended supply-changing path. | One canonical monetary authority; signed policy gates; privilege separation; architecture tests. |
| Economic replay     | One event is repackaged into multiple claims or consumed repeatedly.                       | Canonical event identity; duplicate fingerprints; claim-consumption locks; replay tests.         |
| Oracle dependence   | Many apparent sources share one upstream origin.                                           | Source-lineage graph; source-class independence; dependency scoring.                             |
| Sybil amplification | One participant or machine appears as many economic actors.                                | Canonical participant/asset identity; credential evidence; attestation and fraud controls.       |
| Privacy overreach   | Awareness becomes a centralized personal dossier.                                          | Minimization; purpose limitation; pseudonymity; computation-in-place; access audit.              |
| Policy ambiguity    | Latest methodology silently reinterprets historical events.                                | Versioned policy roots; effective dates; historical replay under original policy.                |
| Environment leakage | Sandbox keys, claims or governance artifacts become valid in production.                   | Environment-scoped key classes, namespaces, feature gates and genesis separation.                |
| State divergence    | Wallet, exchange or database projections drift from canonical protocol state.              | Reconciliation, snapshots, restart tests, deterministic recovery and invariant checks.           |

## Engineering readiness does not equal production authorization

A coherent architecture can still be unfit for public or regulated operation. The system therefore separates architecture completion, software qualification, production configuration, governance authorization and legal/regulatory authority. These are independent gates. A successful technical build should mean the system is ready for controlled qualification and external audit, not that public mainnet issuance or regulated financial functions are automatically authorized.

- Provider and licensing gate: regulated activities require an identified legal and service-provider basis for each jurisdiction.
- Security gate: independent review, key-management controls, penetration testing, recovery exercises and production infrastructure evidence.
- Economic-policy gate: real issuance methodology, supply policy, conversion parameters and governance thresholds must be explicitly approved.
- Operational gate: monitoring, reconciliation, incident response, rollback/abort procedures and evidence retention must be tested.
- Human-governance gate: high-consequence policy and monetary actions require named authority, quorum rules and non-AI approvals where specified.

### Production principle

Code can create a capability. Only governance, legal authority, security evidence and operational controls can authorize its use.

## Where the architecture may be distinctive

The individual ingredients in this white paper are not claimed as new in isolation. Federated databases, knowledge graphs, identity systems, policy engines, oracles, blockchains, cryptographic commitments and privacy controls all have established prior art. Potential technical differentiation lies in the combination, ordering and explicit authority separation used to convert heterogeneous economic evidence into proof-bound sovereign monetary state.

| Potential claim cluster                        | Distinctive combination                                                                                           |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Dual economic ontology on one sovereign ledger | Distinct human-contribution and productive-contribution proof/valuation paths with shared final settlement.       |
| Information vs. Monetary Consensus             | Factual sufficiency is resolved separately from monetary authorization and finality.                              |
| Canonical claim monetization firewall          | Event identity, lineage, duplicate clustering and one-time claim consumption prevent repeated monetary use.       |
| Evidence / Rights / Policy commitments         | Supply-changing events can be cryptographically tied to why they were supported, authorized and interpreted.      |
| Rights-first Human Economy                     | Authorized contribution is valued without requiring generalized human-worth scoring or public raw sensitive data. |
| Source-lineage-aware Productive Oracle Mesh    | Corroboration evaluates upstream independence rather than counting endpoints.                                     |
| Valuation / supply / price separation          | PEVE and GPUV remain distinct from native-asset quantity and exchange price.                                      |
| AI subordinate to monetary governance          | AI can reason over evidence but cannot self-authorize monetary state changes.                                     |

### Patent caution

This is an engineering novelty hypothesis, not a patentability opinion. Formal claims require dedicated prior-art searching and legal analysis across federated data, oracle, provenance, identity/consent, token issuance, data-market and machine-economy systems.

## A monetary system that can explain why state changed

The most important shift in SunRey's Economic Awareness architecture is conceptual. A conventional token system begins with an asset and asks how it should be transferred. SunRey begins with a sequence of prior questions: What economic fact occurred? What evidence supports it? Who or what did it involve? Was the evidence authorized for this purpose? Is the event unique? What methodology interprets it? What governance permits it to influence monetary state? Only then does the protocol ask whether a state transition is valid and final.

This sequencing allows human and autonomous productive activity to remain distinct economic domains while sharing one sovereign proof and settlement architecture. Human contributions can be economically recognized without reducing people to intrinsic scores or publicly exposing sensitive source data. Productive output can be standardized without allowing sensors, telemetry or market prices to become mint commands. External information can expand indefinitely while the monetary core stays narrow.

### Final thesis

SunRey can function as an economic operating system only if it preserves three boundaries: information is not money; valuation is not monetary policy; and intelligence - human or artificial - is not sovereign authority.

The phrase "Economic facts outside the chain. Monetary truth inside the chain." is therefore more than a slogan. It is the architecture's governing separation of concerns: reality is observed and proven through federated, rights-aware evidence systems; monetary truth is finalized only through a governed sovereign protocol capable of proving not merely that state changed, but why.

## REFERENCES

## Historical sources and fact-check basis

- [1] Defense Advanced Research Projects Agency, Director testimony, March 27, 2003. Describes TIA as an experimental, multi-agency prototype network integrating collaboration/decision support, language translation, and data search/pattern-recognition technologies. [Source](#)
- [2] U.S. Department of Defense / DARPA, Report to Congress Regarding the Terrorism Information Awareness Program, May 20, 2003. Primary source for TIA, Genisys, Genisys Privacy Protection and EELD technical descriptions. [Source](#)
- [3] Defense Advanced Research Projects Agency, FY 2004 RDT&E Budget Item Justification, February 2003. Includes Genisys program plans on database federation and schema abstraction. [Source](#)
- [4] Department of Defense Office of Inspector General, Terrorism Information Awareness Program, Report D-2004-033, December 12, 2003. Reviews TIA privacy safeguards, Genisys Privacy Protection and the 2003 congressional funding/termination action. [Source](#)
- [5] DARPA, DARPA: 60 Years 1958-2018. Institutional retrospective noting the TIA program's controversy and termination. [Source](#)

### Source interpretation note

Historical descriptions above are limited to what the cited primary and oversight materials support. References to SunRey architecture, Economic Awareness Fabric, Information Consensus, Canonical Economic Claims, PEVE, GPUV and the five-root proof structure are SunRey design constructs, not claims that those concepts were present in the historical DARPA programs.

## GLOSSARY

## Terminology

| Term                            | Definition                                                                                                                                                                        |
|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Economic Awareness Fabric (EAF) | Federated, rights-aware layer for provider integration, normalization, provenance, entity resolution, graph relationships and evidence sufficiency. It is not monetary authority. |
| Economic Observation Envelope   | Canonical representation of a provider observation that preserves source identity, time, assertion, units, rights, provenance and quality state.                                  |
| Information Consensus           | SunRey evidence-sufficiency mechanism answering whether authorized evidence is sufficient to support an economic fact. It is distinct from monetary finality.                     |
| Verified Economic Fact          | A proposition supported under an applicable evidence policy. It may still be ineligible for monetary use.                                                                         |
| Canonical Economic Claim        | Durable economic proposition with canonical event/entity identity, lineage, evidence, rights, policy, challenge and monetization state.                                           |
| Economic Knowledge Graph        | Semantic relationship layer connecting entities, assets, contributions, events, evidence, rights and claims.                                                                      |
| PEVE                            | Personal Economic Value Engine; a versioned valuation layer for eligible human contribution. It does not assign intrinsic human worth and cannot mint.                            |
| GPUV                            | Gross Productive Unit Value; a standardized productive-value representation downstream of verified productive contribution and upstream of monetary policy.                       |
| Monetary Consensus              | Protocol validation/finality plane for authorized monetary state transitions. It does not independently establish external facts.                                                 |

## Proof commitments and terminology discipline

| Term                | Definition                                                                                                                                      |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidence Root       | Cryptographic commitment to the evidence supporting a monetary event.                                                                           |
| Rights Root         | Cryptographic commitment to the relevant rights, consent and authorization context.                                                             |
| Policy Root         | Cryptographic commitment to the exact policy/methodology version governing the transition.                                                      |
| Transaction Root    | Cryptographic commitment to the set of executed transactions or state-changing operations.                                                      |
| Monetary State Root | Cryptographic commitment to resulting canonical monetary state.                                                                                 |
| Claim consumption   | Durable state indicating that an eligible claim has already been used in a governed monetary process, preventing replay or double monetization. |

### Terminology discipline

"Consensus" is used in two distinct senses in this architecture. Information Consensus is an evidence-sufficiency construct; Monetary Consensus is protocol finality. They should never be conflated in implementation, governance or public communication.

Likewise, the five Roots are architectural commitments to execution, state, evidence, rights and policy context. Their exact cryptographic primitive is implementation-dependent; the publication claim is the separation and binding of proof domains, not a mandate for a specific tree or hash construction.